Scraping and Analyzing NCAA Volleyball Data

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1. Motivation

- NCAA Volleyball data availability for instructors and analysts
  - ➢ NCAA has standardized stats pages
- Easy way to grab and compare different seasons
- Create a storage design for analyzing
- Replicated for different teams, divisions, or sports

2. Getting the Data

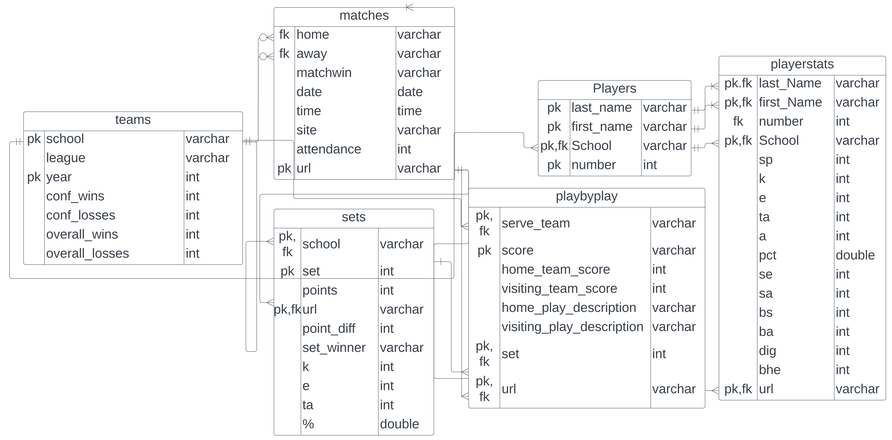
1. Get list of URLs for each school
   a = **https://saintsathletics.com/**
2. Loop to attach common identifier, and desired year(s)
   b = "sports/womens-volleyball/stats/" + /2021
3. Loop to Get Every Match URL (add match ID)
   c = **/Nazareth-college/boxscore/22685**

a+b+c = https://saintsathletics.com/sports/womens-volleyball/stats/2021/nazareth-college/boxscore/22685

4. Loop over all Match URLs to:
   1. Scrape for all data
   2. Clean/Format data
   3. Save into one file

About the Data:
- ➢ All 2013 through 2022 seasons
- ➢ Liberty League
- ➢ Match URL is the key

3. ER Diagram



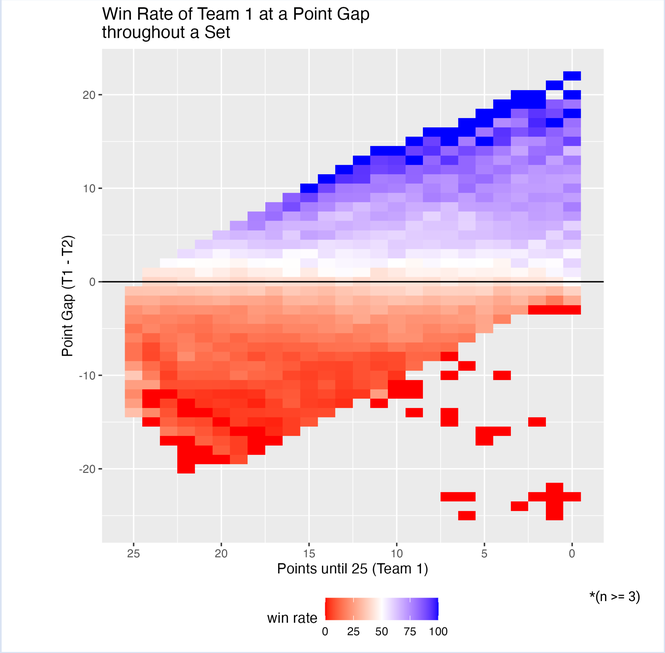
Visual representation of database table relationships.

4. Analyzing the Data

A. How Win Rate %'s change throughout a match
- ➢ Is there a point in the set where the point gap is large enough to consider it over?

B. How various metrics/statistics in the game contribute to and effect whether a team is successful
- ➢ What kind of players are most important & see if serves, kills, or digs, etc. contribute more to winning?

4a. At what point in the game, and at what point gap can you safely say team one will win?



- Need a point gap of close to 10 points to be able to effectively call a match
- Chances of a team winning/losing is fairly consistent throughout the set
- No clear home team advantage
  - ➢ Team 1 is the home team here unless it was arbitrarily assigned (tournaments)

4b. What's the best predictor for if a team will win the match?

Logistic Regression on Volleyball Statistics' effect on Winning a Match

volleyball stat	estimate	test statistic	pvalue
(Intercept)	-2.472	-11.182	< .001
kills	0.150	7.800	< .001
errors	-0.088	-8.134	< .001
total_attempts	-0.044	-10.520	< .001
assists	-0.016	-0.830	0.407
serve_aces	0.189	13.440	< .001
serve_errors	-0.046	-2.890	0.004
dig	0.042	8.140	< .001
solo_blocks	0.140	4.470	< .001
block_assists	0.030	3.146	0.002

- Kills and Blocks positively contribute to wins and Errors negatively contribute, as we would expect to see
- Total attempts is notably negative, putting emphasis on quality over quantity of hits/attempts
- Serve Aces are the most impactful predictor, whereas serve errors only marginally reduce likelihood of winning

4c. Future Analysis

- Serve Errors/Aces Impact throughout a Match
  - ➢ Player level, Aces come from good server
  - ➢ Offensively vs Defensively
- Team compositions & Types of Players
- Look at other divisions and/or non-collegiate teams and analyze differences between them